

Advancing Pharmaceutical Distribution Through Data-Driven Inventory Optimization, Demand Forecasting, and Retailer Performance Analytics

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title “**Advancing Pharmaceutical Distribution Through Data-Driven Inventory Optimization, Demand Forecasting, and Retailer Performance Analytics**”. I extend my appreciation to **Sevam Pharma LLP**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

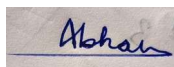
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:

A rectangular box containing a handwritten signature in blue ink that reads "Abhay".

Name: Abhay Sharma

Date: 13-02-2026

Executive Summary:

- ***About the Organization:***

The business that I have collected data from is Sevam Pharma LLP, a pharmaceutical distribution firm based in Sangli, Maharashtra. The company operates in the B2B segment and supplies medicines to retail medical stores in the region, primarily sourcing products from Sun Pharma.

- ***About the problem:***

The major challenges the business is facing include difficulty in managing inventory, which leads to losses due to product expiry, overstocking of slow-moving medicines, and stockouts of fast-moving items. The organization also struggles to track inconsistencies in historical stock and sales records, making it difficult to take timely decisions. This results in increased manual effort and money being blocked in inventory. In addition, the business lacks clear visibility into distribution across retailers, due to which expected sales are not achieved even when demand is high.

- ***Possible Approaches and Solutions:***

By analyzing recent sales data, demand patterns and fluctuations can be identified to support better decisions on when and how much stock to procure. This can help reduce excess inventory and money blocked in stock. Further, studying order frequency and demand trends at both the product and retailer levels will help identify slow-moving items, fast-moving items, and retailers that contribute more to overall profitability.

Organization Background:

Sevam Pharma LLP is a pharmaceutical distribution firm registered as a Limited Liability Partnership with the Registrar of Companies in Pune. It was incorporated on 13 January 2022 and operates in Sangli, Maharashtra. The company is primarily engaged in wholesale and trading activities, with a focus on the distribution of pharmaceutical products and related non-durable goods in the regional market. The organization is led by a team of designated partners including Ajit Anna Khavate, Shridhar Babasaheb Dhamnkar, Avinash Krishnaji Pore, Sandeep Ashokrao Patil, and Sachin Dhanchandra Sakale, who have been associated with the company since its inception. Sevam Pharma works closely with Sun Pharma for medicine procurement and supplies medicines to retail medical stores across its operating region. The company largely distributes medicines used in the treatment of neurological and psychiatric conditions such as epilepsy, Parkinson's disease, depression, anxiety disorders, schizophrenia, and other central nervous system (CNS)-related conditions, ensuring the availability of essential therapies to its retail partners.

Problem Statements:

- 1.Improving warehouse inventory and expiry management using historical stock and sales data to reduce losses and improve profitability.
- 2.Analyzing the reasons for poor performance of specific retail outlets in terms of sales and inventory levels.
- 3.Identifying seasonal and regional demand patterns to improve forecasting and stock allocation across retailers.
- 4.Understanding the causes of anomalies and inconsistencies in stock and sales data using historical records.

Background for Problems

Following discussions with management and observation of operations, it was found that challenges at Sevam Pharma LLP developed gradually as the business expanded and the number of retail outlets increased. The warehouse began handling a wider range of medicines, many with limited shelf life, without a matching improvement in inventory planning practices. The major cause was the lack of a structured, data-driven approach to demand forecasting and expiry-focused inventory management, which led to overstocking of slow-moving medicines and stockouts of fast-moving products.

Internal Factors Leading to the Cause:

- Manual reconciliation of stock and sales records, causing errors and delays
- Limited visibility into retailer-wise demand and inventory movement
- Inefficient route planning and delivery scheduling, increasing logistics costs
- Inconsistent data across reports and systems

External Factors Contributing to the Issue:

- Seasonal and regional variations in medicine demand
- Variability in supplier lead times and transportation delays
- Market-level demand fluctuations not captured in planning

These combined factors have increased operational effort, caused expiry-related losses, and led to frequent stock imbalances. As a result, working capital remains blocked in inventory, service levels to retailers are inconsistent, and distribution costs have increased, affecting overall efficiency and profitability.

Problem Solving Approach:

- 1) For the First Problem (**Inventory & Expiry Management**):
 - a) As the business maintains data related to sales, purchases, current stock, supplier details, and expiry dates for medicines, this data can be used for analysis.
 - b) With the help of the available data, useful insights can be extracted, such as:
 - i) Identifying sales trends and detecting outliers or unusual spikes/drops in sales.
 - ii) Understanding which products and therapeutic categories are fast-moving and which are slow-moving.
 - iii) Analyzing month-wise demand patterns to understand seasonal variations in sales.
 - c) Using these insights, inventory levels can be better planned, expiry risk can be reduced, and stock procurement can be optimized.
- 2) For the Second Problem (**Retail Outlet Performance Analysis**):
 - a) By using outlet-wise sales and stock data provided by the distributor, performance differences across retailers can be analyzed.
 - b) Key analyses include:
 - i) Comparing sales velocity with inventory holding to identify underperforming outlets.
 - ii) Studying outlet-wise sales on a weekly/monthly basis to observe consistent low or high performance.
 - iii) Identifying gaps in stock allocation that may be affecting sales at certain outlets.
 - c) These insights can help in recommending corrective actions such as adjusting stock allocation and delivery frequency.
- 3) For the Third Problem (**Seasonal & Regional Demand Patterns**):
 - a) Historical sales data across months and retailers can be analyzed to identify seasonal demand trends.
 - b) This includes:
 - i) Identifying products with seasonal spikes in demand.
 - ii) Comparing demand across different areas served by the warehouse to identify regional variations.
 - iii) Using simple forecasting methods to estimate future demand at product and outlet levels.
 - c) These insights can support better forecasting and more balanced stock allocation across retailers.
- 4) For the Fourth Problem (**Root Causes of Anomalies & Data Inconsistencies**):
 - a) Since sales and stock levels naturally fluctuate, not every change can be treated as an issue; however, major deviations can be analyzed.
 - b) When anomalies are observed, the following steps can be taken:
 - i) Checking whether the anomaly is due to internal reasons (data entry errors, stock misreporting, delayed updates).
 - ii) Verifying external reasons such as supply delays, sudden demand changes, or distribution issues.
 - iii) Comparing patterns before and after the anomaly period to understand possible causes.
 - c) This helps in improving data reliability and operational decision-making.
- 5) **Tools:** Excel and Python will be used for data cleaning, exploratory analysis, trend analysis, simple forecasting, and anomaly detection.

Expected Timeline:

i. Work Breakdown Structure:

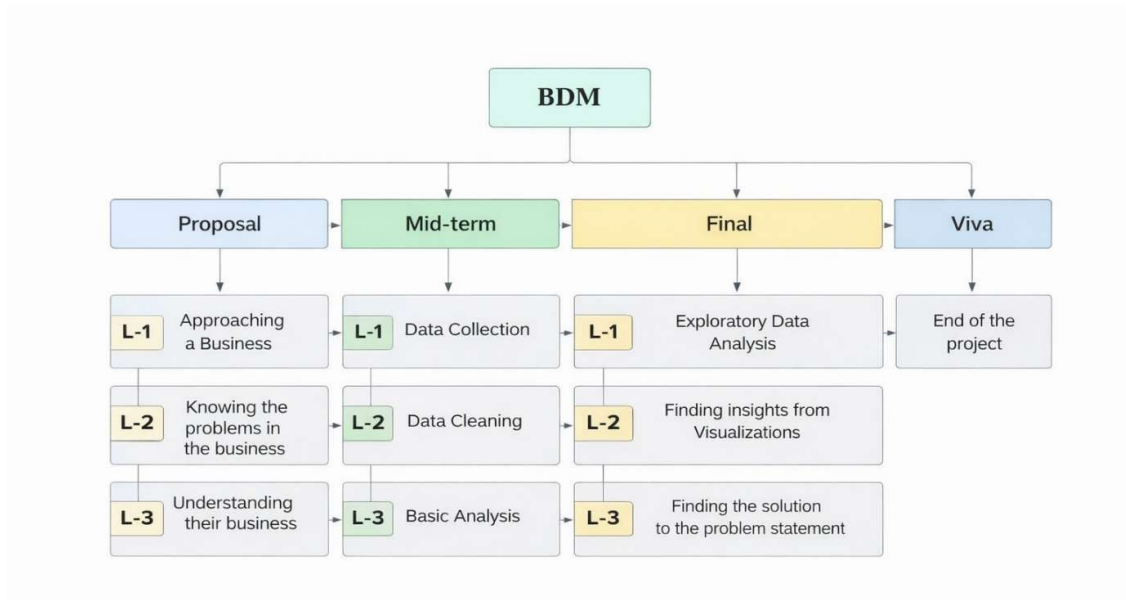


Figure 1 Work Breakdown Structure of BDM Project

ii. Gantt Chart:

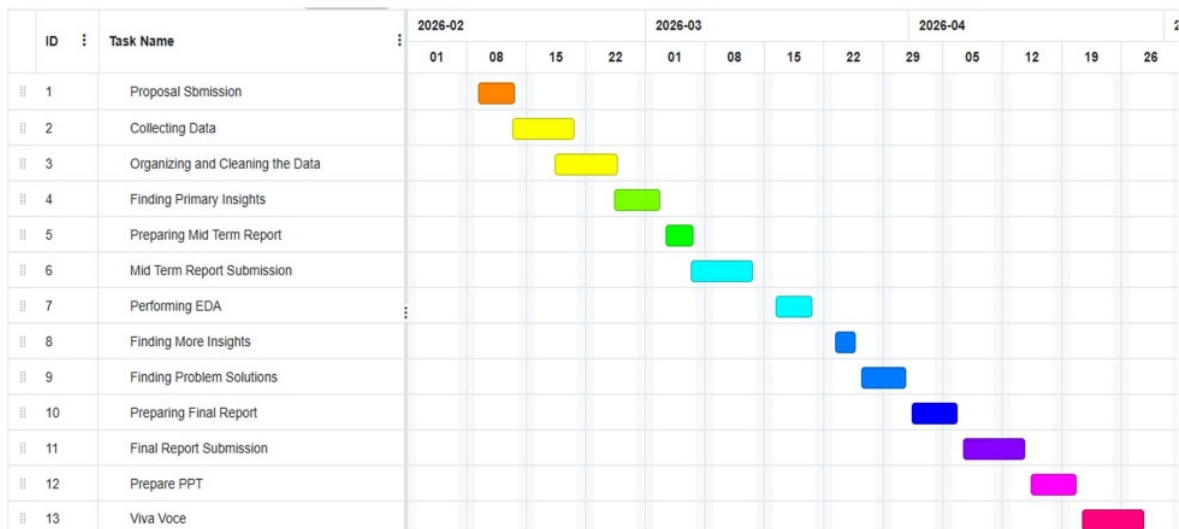


Figure 2 Gantt Chart for expected timeline of project

Expected Outcomes:

The main goal of this project is to identify patterns in stock movement and sales across different months, products, and retail outlets, and to use these insights to support better inventory and distribution decisions. The expected outcomes include:

- **Identification of Seasonal Demand Patterns:**
Analysis of historical sales and stock data is expected to highlight seasonal trends in medicine demand, such as peak periods for specific product categories, supporting better procurement planning.
- **Product-Specific Insights:**
The study is expected to reveal differences in performance across products, helping distinguish fast-moving and slow-moving medicines and identify areas with higher expiry risk.
- **Retailer-Level Performance Insights:**
Outlet-wise sales and inventory analysis is expected to provide visibility into performance differences across retailers and highlight factors contributing to underperformance.

Overall Outcome:

The overall outcome is to enable more balanced stock allocation across retailers and time periods, reduce avoidable stockouts and expiry risks where feasible, and support more data-informed operational decisions across the distribution network.